Modern Guiding Principles in Cryptography

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- **Kerckhoffs' Principle**

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- **Kerckhoffs' Principle**
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- **Trusted Third Parties**
- **Human Factors**

Kerckhoffs' Principle: Security through obscurity = BAD!

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Secure distribution of key material does not scale well to large networks

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  - More difficult to detect a compromise

Public key cryptography addresses the key distribution problem

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- **Do users really need a shared secret?**

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- Distribution with TRUST much more complicated

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- **Trent distributes secret symmetric keys**

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    - Trusted to verify the owner of a key to others

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In Review – Core principles and issues

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- Symmetric key management is critical – and hard
- Asymmetric keys address key distribution problem
- Trusted third parties play a crucial role
- Human factors always at play